ME-OPS v2 — Forensic Technical Handoff

Operator-Grade Takeover Package

Audit Date: 2026-03-21 | **Machine:** GettUppENT | **User:** finan | **Auditor:** Claude Sonnet 4.6

⚠️ CRITICAL SECURITY NOTICE — READ FIRST

TWO GEMINI API KEYS ARE EXPOSED IN PLAINTEXT IN SOURCE FILES:

- `AIzaSyBcXkir3PO1YoNIzYnINv35yzZopUDdzes` — **Hardcoded** in `run_phase2.py:6` and `core/recall/gemini_embed.py:17` and `_test_embed.py:6`
- `AIzaSyAnLCFiQPIZs_AlEPCBqSmjlVvLgSqlQM4` — **In shell environment** (set separately, used by `_audit2.py`)

These keys are different. The environment key overrides the source key at runtime only if set before script start. There is no `.env` file. No secrets management. Move both keys out of source immediately before sharing anything.

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1. EXECUTIVE SUMMARY

Field	Value
Project Name	ME-OPS v2
Purpose	Personal Causal Intelligence System — turns raw digital history into failure chains, win signatures, heuristics, and operator briefings
Intended Operator	Johnny Cage (solo founder, Minneapolis)
Real MVP Goal	Answer 8 evidence-based questions about personal work patterns from lived digital history
Current Maturity	Phase 3 of 5 complete. Functional but fragile.
Usable Today?	**YES** — `python run_phase2.py recall <query>` produces real semantic recall. Core pipeline runs end-to-end.
DB Status	2.07 MB SQLite. 561 entities, 579 embeddings, 207 sessions, 18 cases, 10 heuristics. Confirmed live.

Top Blockers

1. **311/526 events (59%) have NULL project_id** — unclassified. Core intelligence degrades proportionally.
2. **0 decisions extracted** — `decision_replay.py` extracts nothing despite 526 events scanned.
3. **0 context collapse alerts** — detector logic scores all sessions below threshold.
4. **No test suite** — zero test coverage across all modules.
5. `requirements.txt` **is wrong** — missing `google-genai` and `numpy`. Fresh install fails silently.

Top Risks

1. **Two exposed Gemini API keys in source** (CRITICAL)
2. `pieces_bridge/ingest.py` **appears active but is fully dead** — misleads any new maintainer
3. **No DB backup** — single SQLite file, one mistake = full data loss
4. `_reset_canonical.py` **is a silent data destruction script** — no confirmation prompt

Immediate Next Actions

1. Rotate both Gemini API keys, move to `.env`
2. Backup `meops.db` NOW
3. Fix `requirements.txt`
4. Mark `ingest.py` as deprecated
5. Run `python run_phase2.py stats` to confirm current state

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2. PROJECT INTENT vs CURRENT REALITY

Intended Architecture

```
PiecesOS (localhost:39300)
    ├── WebSocket stream → real-time summary IDs
    └── REST API → full content fetch
         └── ME-OPS Ingestion
              └── Canonical Ledger (SQLite)
                   ├── Cases (failure chains, win signatures, decision replays)
                   ├── Heuristics (auto-mined operating laws)
                   ├── Embeddings (768d Gemini vectors, cosine search)
                   └── Operator Briefings + Live Guardrails
```

Actual Implementation (Confirmed)

```
PiecesOS (localhost:39300)
    └── MCP SSE endpoint ONLY (REST /workstream_summaries/{id} returns 404)
         └── mcp_client.py → batch_snapshot via SSE+JSON-RPC
              └── ingest_p2.py → local cache re-classification + MCP delta
                   └── SQLite (meops.db)
                        ├── 526 events (311 NULL project, 215 classified)
                        ├── 26 failures + 9 outcomes (lifted from events)
                        ├── 9 failure chains + 9 win signatures
                        ├── 579 Gemini 768d embeddings (cosine search working)
                        └── 10 manually seeded heuristics (auto-mining NOT built)
```

Architecture Gaps

Intended	Actual	Status
WebSocket real-time ingestion	REST polling + local cache	Dead path (ingest.py)
Decision extraction from history	0 extracted despite 526 events	Broken (wrong trigger phrases)
Context collapse detection	0 alerts from 207 sessions	Broken (scoring threshold logic)
Auto-mined heuristics (Phase 4)	10 manually seeded	Not built
Live operator guardrails (Phase 5)	Not started	Not built
Pydantic validation on DB writes	Models defined, not enforced	Documentation only
`core/causal/` inference module	`__init__.py` only	Empty scaffold
`pipelines/` standalone scripts	`__init__.py` only	Empty scaffold

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3. MVP DEFINITION

Minimum Viable Workflow (Confirmed Working Today)

```
cd C:\Users\finan\me-ops
set PYTHONIOENCODING=utf-8
set GEMINI_API_KEY=<key>
python run_phase2.py full        # ingest → sessionize → cases → embed
python run_phase2.py recall "IronClad revenue blocked"   # semantic search
python run_phase2.py brief       # operator briefing
```

MVP Acceptance Criteria

- [x] Ingest PiecesOS history into canonical DB
- [x] Build sessions from event stream
- [x] Lift failures and outcomes from event text
- [x] Build failure chains per project
- [x] Embed all entities with Gemini (768d)
- [x] Semantic recall returns ranked prior cases with similarity > 0.70
- [ ] Decision extraction working (0/526 extracted — FAILING)
- [ ] Context collapse alerts firing (0/207 sessions — FAILING)
- [ ]

  > *50% events classified to specific projects (41% now — FAILING)*

Out of Scope for MVP

- GUI/dashboard
- Autonomous agent integration
- Multi-user support
- Cloud sync or remote DB
- Real-time WebSocket streaming (polling is sufficient for MVP)
- Auto-mined heuristics (Phase 4)
- Live guardrails (Phase 5)

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4. COMPLETE FILE AND FOLDER INVENTORY

Root Level

File	Status	Lines	Purpose	Safe to Edit?
`run_phase2.py`	**PRIMARY ENTRY POINT**	167	Full Phase 2+3 pipeline. ALL production commands route through here.	Yes
`run.py`	Superseded	119	Original Phase 1 runner. `stats`, `sync`, `cases`, `brief`, `seed` commands still work but don't include Phase 2+ logic.	Yes — but prefer run_phase2
`requirements.txt`	**WRONG — FIX IMMEDIATELY**	2	Only declares `pydantic>=2.0` and `websockets>=12.0`. Missing `google-genai` and `numpy`. Fresh install breaks without them.	Yes — must fix
`meops.db`	**PRECIOUS — BACKUP NOW**	—	2.07 MB SQLite. Single source of all canonical data. 526 events, 579 embeddings, 207 sessions. Not in git.	Never edit manually
`_reset_canonical.py`	**DANGEROUS**	23	Wipes all derived entities, cases, sessions. Resets raw cache processed flag. No confirmation prompt.	No — label "DESTRUCTIVE" first
`_test_embed.py`	Keep as health check	16	Tests Gemini API key + model endpoint. Safe quick check. Hardcodes key `AIzaSyBcXkir3PO1YoNIzYnINv35yzZopUDdzes`.	Yes
`_debug.py`	Safe to delete	23	One-off diagnostic for annotation_text sampling. No longer needed.	Yes — delete
`_enc.py`	Safe to delete	3	One-off encoding test.	Yes — delete
`_migrate.py`	Keep for reference	33	Ran once to add `fragmentation_score` to `context_metrics`. Migration already applied. Safe to delete after confirming schema.	Yes — delete after review
`_audit.py`, `_audit2.py`, `_audit3.py`	Scratch — delete	60-83	Generated during this audit.	Yes — delete

core/storage/

File	Status	Purpose	Safe?	Risk
`schema.py`	**LOAD-BEARING**	Full SQLite DDL + `SEED_PROJECTS_SQL`. Source of truth for DB structure. Contains `unknown` project seed. `context_metrics.fragmentation_score` column added via ALTER (now in CREATE).	Dangerous	Schema change requires migration + potential data wipe
`db.py`	**LOAD-BEARING**	`get_conn()`, `init_db()`, `EntityRepo` with `upsert()`, `ensure_project()`, `add_edge()`. DB path defaults to `Path.home() / "me-ops" / "meops.db"` — overridable via `MEOPS_DB` env var.	Dangerous	All ingestion breaks if `upsert()` signature changes

Key behavior (Confirmed): `EntityRepo.upsert()` uses `ON CONFLICT(id) DO UPDATE` — safe for re-runs. `ensure_project()` auto-creates project rows to satisfy FK constraints.

core/ledger/

File	Status	Purpose	Safe?	Key Quirk
`classify.py`	**High-impact**	Maps text → `project_id` (12 project buckets) and `event_kind`. Rule-based keyword scoring. Controls classification quality.	Yes — add keywords freely	Adding wrong signals causes mis-classification. 311/526 events return NULL (no match).
`sessionize.py`	Active	Groups events by 30-min gap / 4-hour max. Computes `tool_switch_count`, `fragmentation_score`, `context_load_score`.	Yes	`fragmentation_score` is based on distinct tool count per session — NOT real app-switch events. All sessions score ≤0.12 fragmentation.

Confirmed Gap: `classify.py` returns `None` (stored as SQL NULL) for unmatched text — not the string `"unknown"`. The `unknown` row in projects table exists but is never used by classify. The prior handoff doc saying "311 stored as `project_id='unknown'`" was **wrong** — they are NULL.

core/cases/

File	Status	Lines	Output	Safe?
`failure_chain.py`	Active	163	`lift_failures_and_outcomes()` → 26 failures + 9 outcomes. `build_failure_chains()` → 9 chains.	Yes
`win_signature.py`	Active	130	`build_win_signatures()` → 9 win signatures from outcomes.	Yes
`decision_replay.py`	**Broken — 0 output**	151	`extract_decisions()` → 0 decisions from 526 events despite 56 trigger phrases.	Yes

core/recall/

File	Status	Lines	Purpose	Safe?	Key Config
`gemini_embed.py`	**Phase 3 core**	342	Full Gemini pipeline: batch embed → SQLite blob → cosine search → format report. Rate limit: 100 RPM free tier, batch 50, 65s delay, retryDelay respected. **Contains hardcoded API key.**	Yes — carefully	BATCH_SIZE=50, BATCH_DELAY=65s, EMBED_DIMS=768, model=gemini-embedding-2-preview
`retrieve.py`	Active fallback	257	TF-IDF index with cosine sim. Used when Gemini quota hit (embed_query returns None).	Yes	In-memory index rebuilt each call (not cached). Slower than Gemini but works offline.

Confirmed working (2026-03-21): Recall query "IronClad ZIP API revenue blocked deployment" returns results with similarity 0.74–0.76 using Gemini embeddings.

core/heuristics/

File	Status	Purpose	Safe?
`seed.py`	Active	10 manually crafted heuristics. Seeded once via `init_db()` → `seed_heuristics()`. Re-runs are idempotent (INSERT OR IGNORE).	Yes — add more heuristics freely

core/briefing/

File	Status	Lines	Purpose	Safe?
`daily.py`	Active — template only	217	3 briefing templates: daily, incident, project. **Hardcoded narrative content.** Not derived from case analysis. DB stats injected but narrative is static.	Yes

core/operator/

File	Status	Lines	Output	Safe?
`collapse_detector.py`	Active — 0 output	111	Scans 207 sessions, scores collapse risk, writes interventions. **Currently produces 0 alerts.** Threshold 0.6 never reached because `tool_switch_count` is distinct-tools count (max 1 per single-tool session), not real switching events.	Yes

core/models/

Files	Status	Purpose	Safe?
`base.py`, `event.py`, `decision.py`, `case.py`	**Documentation only**	Pydantic models defining the canonical object schema. NOT validated on DB writes. `EntityRepo.upsert()` accepts raw dicts and stores type-specific fields in a `data` JSON blob.	Yes — models do not affect runtime

core/causal/

Status	Content	Notes
Empty scaffold	`__init__.py` only	Phase 4 causal inference. Not started. Safe to ignore.

pieces_bridge/

File	Status	Lines	Safe?	Critical Note
`ingest.py`	**DEAD — DO NOT USE**	228	Docstring claims "live ingestion bridge." `pieces_get()` calls REST endpoint `/workstream_summaries/{id}` which returns HTTP 404 for all IDs. WebSocket path exists but calls a dead endpoint. **Appears functional but silently fails.**	Do not call
`ingest_p2.py`	**Active — primary ingestion**	164	Re-classifies from local raw cache (526 items) + fetches delta from MCP. Works.	Yes
`mcp_client.py`	**Active — critical dependency**	101	Pure MCP over SSE. Opens session, sends JSON-RPC batch_snapshot calls. Used by `ingest_p2.py`. Port 39300 hardcoded.	Dangerous

tmp/pdfs/

File	Status	Lines	Notes
`generate_me_ops_app_summary.py`	**Unknown — not wired**	230	Not imported by anything. Appears to generate PDF documentation. Not referenced in run scripts. Safe to ignore. Status: Likely (orphaned)

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5. VERSIONS AND DEPENDENCY MATRIX

Component	Confirmed Version	Location	Declared in requirements.txt?
Python	**3.14.2**	`C:\Python314\python.exe`	No
google-genai	**1.68.0**	User site-packages	**NO — MISSING**
pydantic	**2.12.0**	User site-packages	Yes (>=2.0)
numpy	**2.4.3**	User site-packages	**NO — MISSING**
websockets	**16.0**	User site-packages	Yes (>=12.0)
node.js	v22.22.1 (nvm)	`C:\Users\finan\AppData\Local\nvm\`	No — external dependency
mcp-remote	0.1.38	npm global	No — external dependency
PiecesOS	**12.3.10**	Windows system tray	No — external service
SQLite	stdlib (3.x)	Python stdlib	No

Correct `requirements.txt` (must fix):

```
pydantic>=2.0
google-genai>=1.68.0
numpy>=2.0
websockets>=12.0
```

External Dependencies (not in any manifest)

- **PiecesOS 12.3.10** — must be running on port 39300 before any ingestion
- **LTM-2.7 engine** — must be enabled in PiecesOS settings
- **Node.js v22+** — for MCP test scripts only; not required for main pipeline
- **mcp-remote 0.1.38** — for Claude Desktop MCP; not required for main pipeline

Dependency Risks

- **Python 3.14** is cutting-edge (Dec 2025). Some packages may not fully support it. `pydantic-core` compiled wheels confirmed working. Risk: Low but real for any new pip installs.
- **google-genai 1.68.0** uses `tenacity` for retries — this is what catches 429 errors and re-raises them. The custom retry logic in `gemini_embed.py` intercepts these correctly.
- **No lockfile** — pip will install latest compatible on fresh install. Version drift risk is real.

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6. MACHINE AND ENVIRONMENT PROFILE

Core Machine Facts (Confirmed)

Item	Value	Confirmed?
OS	Windows 11	Confirmed
Machine name	GettUppENT	Confirmed (in WSL hostname)
User	finan	Confirmed
Shell (default)	PowerShell	Confirmed — cmd.exe used for some commands
Python path	`C:\Python314\python.exe`	Confirmed
User site-packages	`C:\Users\finan\AppData\Roaming\Python\Python314\site-packages\`	Confirmed
DB default path	`C:\Users\finan\me-ops\meops.db`	Confirmed (Path.home() / "me-ops" / "meops.db")
MEOPS_DB env var	NOT SET	Confirmed — uses default path
PiecesOS port	39300	Confirmed live
WSL2 present	Yes	Confirmed (used by Paperclip/OpenClaw stack)
nvm	v22.22.1	Confirmed

Environment Variables

Variable	Required?	Where Set	Where Used	Current Value
`GEMINI_API_KEY`	**YES**	Shell environment	`gemini_embed.py:17` (falls back to hardcoded key if not set)	`AIzaSyAnLCFiQPIZs_AlEPCBqSmjlVvLgSql` in shell
`PYTHONIOENCODING`	YES on Windows	`run_phase2.py:5` sets it	All scripts — prevents charmap errors	Set by run_phase2.py at startup
`MEOPS_DB`	No	Not set	`db.py` — overrides default DB path	Not set; default path used

Critical: TWO DIFFERENT API KEYS

- **In shell environment:** `AIzaSyAnLCFiQPIZs_AlEPCBqSmjlVvLgSqlQM4`
- **Hardcoded in source:** `AIzaSyBcXkir3PO1YoNIzYnINv35yzZopUDdzes`

`gemini_embed.py:17` uses `os.environ.get("GEMINI_API_KEY", "hardcoded_key")` — env var takes priority. So at runtime, the env key is used. But if env var is not set on a fresh machine, the hardcoded key is used. Both are exposed. **Rotate both.**

Windows-Specific Behavior

- **PowerShell `-c` with Python**: Inline Python strings with quotes fail in PowerShell. Use `.py` files.
- **`charmap` codec error**: Any script printing non-ASCII (→, ✅, emoji) crashes without `PYTHONIOENCODING=utf-8`. `run_phase2.py` sets this at startup. Other scripts do not.
- **Path separators**: Project uses raw strings (`r'C:\...'`) correctly. No forward-slash issues.
- **Port 39300**: Hardcoded in `mcp_client.py` and `ingest.py`. PiecesOS port can vary per install — check Quick Menu if ingestion fails.

Machine-Specific Hacks (Confirmed)

1. `unknown` project row manually inserted mid-session to unblock FK constraint failures. Now in `SEED_PROJECTS_SQL` so fresh installs include it.
2. `context_metrics.fragmentation_score` added via ALTER TABLE (`_migrate.py`) after initial schema deployment. Column now in `SCHEMA_SQL` CREATE statement — fresh installs are fine.
3. `PYTHONIOENCODING=utf-8` must be set in every script or as env var on Windows — not handled globally.

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7. SETUP AND RECOVERY RUNBOOK

Cold Start — New Machine

```
# 1. Prerequisites
# - Install Python 3.12+ (3.14 confirmed working)
# - Install PiecesOS from https://pieces.app
# - Enable LTM-2.7 engine in PiecesOS Desktop App
# - Obtain Gemini API key from https://aistudio.google.com

# 2. Copy project
# Copy entire C:\Users\finan\me-ops\ to new machine

# 3. Fix and install dependencies
cd me-ops
# Edit requirements.txt to:
# pydantic>=2.0
# google-genai>=1.68.0
# numpy>=2.0
# websockets>=12.0
pip install -r requirements.txt

# 4. Create .env file (DO THIS FIRST)
# Create C:\Users\{you}\me-ops\.env containing:
# GEMINI_API_KEY=your_key_here
# MEOPS_DB=C:\Users\{you}\me-ops\meops.db  (optional)

# 5. Set env vars before any run
set PYTHONIOENCODING=utf-8
set GEMINI_API_KEY=your_key_here

# 6. Initialize DB (creates schema + seeds projects + heuristics)
python run_phase2.py full

# 7. Verify
python run_phase2.py stats
python run_phase2.py recall "IronClad revenue blocked"
```

Minimal Known-Good Boot

```
cd C:\Users\finan\me-ops
set PYTHONIOENCODING=utf-8
set GEMINI_API_KEY=<key>
python run_phase2.py stats    # verify DB counts
```

Health Checks

```
# 1. PiecesOS alive?
curl http://localhost:39300/.well-known/version
# Expected: 12.3.10

# 2. Gemini API working?
python _test_embed.py
# Expected: OK: 768 dims

# 3. DB integrity?
python run_phase2.py stats
# Expected: events=526, embeddings=579, sessions=207

# 4. Recall working?
python run_phase2.py recall "connection refused gateway"
# Expected: results with sim > 0.60
```

Safe Reset (Rebuild Derived Layers Only)

```
# BACK UP FIRST
copy C:\Users\finan\me-ops\meops.db C:\Users\finan\me-ops\meops.db.bak

# Reset derived layers (keeps raw_pieces_summaries)
python _reset_canonical.py

# Rebuild everything
set GEMINI_API_KEY=<key>
python run_phase2.py full
```

Disaster Recovery (DB Lost)

```
# Raw cache is in raw_pieces_summaries — IF DB still exists
# If DB completely lost: must re-ingest from PiecesOS
# PiecesOS must be running with LTM enabled

python run_phase2.py full    # fetches delta from PiecesOS MCP
```

Known Bad Commands (Do Not Run)

```
# DANGER: Wipes all derived data without confirmation
python _reset_canonical.py    # only if you intend full rebuild

# DEAD: Will 404 on all IDs, produces bad data
# from pieces_bridge.ingest import PiecesBridge  # DO NOT USE
```

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8. ERROR AND FAILURE HISTORY

#	Error	File	Trigger	
1	`HTTP 404` on all REST summary fetches	`ingest.py`	`pieces_get("/workstream_summaries/{id}")`	Wı en for
2	`sqlite3.IntegrityError: FOREIGN KEY constraint failed`	`failure_chain.py`, `win_signature.py`, `decision_replay.py`	Building cases with `project_id=NULL`	FK cas ref pro pro row
3	`UnicodeEncodeError: 'charmap' codec`	`run_phase2.py`	Printing →, ✅, emoji on Windows default cp1252	Wi en
4	`OperationalError: table context_metrics has no column fragmentation_score`	`collapse_detector.py`	First run after schema was already deployed without the column	Sc de co AL in
5	`AttributeError: 'sqlite3.Row' has no attribute 'get'`	`collapse_detector.py`, `win_signature.py`	Calling `.get()` on sqlite3.Row objects	sq no row row
6	`429 RESOURCE_EXHAUSTED` (Gemini)	`gemini_embed.py`	Batching 100 texts to free-tier API (100 RPM limit)	Fre rec Fir 10 lim
7	`SyntaxError` in PowerShell inline Python	Multiple	`cmd /c "python -c "..."` with nested quotes	Po int qu dif cm
8	`ERR_UNESCAPED_CHARACTERS` in Node.js	`pieces_mcp_test.js`	Token in SSE URL contains `=` chars, passed as raw string to `http.request`	No cli

				UR sp
9	`0 decisions extracted`	`decision_replay.py`	All 526 events scanned	Tri de dia lar wo su pa na
10	`0 context collapse alerts`	`collapse_detector.py`	207 sessions scored	`too` is co ses for ses rea
11	311/526 events NULL project_id	`classify.py`	Events whose text has no matching keyword	Ke co inc WS inf ses
12	Previous handoff doc claimed 1374 raw summaries	`docs/ME_OPS_v2_HANDOFF.pdf`	Stale documentation	DB be an

9. DATABASE AND STATE LAYER

Schema Summary

Table	Rows	Purpose	Precious?	Rebuildable?
`entities`	561	All canonical objects: 526 events, 26 failures, 9 outcomes	Yes	Yes — from raw cache + PiecesOS
`projects`	13	12 venture buckets + "unknown" catch-all	No	Yes — from `SEED_PROJECTS_SQL`
`sessions`	207	Event groupings with fragmentation scores	No	Yes — from events
`cases`	18	9 failure chains + 9 win signatures	Semi	Yes — from failures/outcomes
`case_members`	35	Entity-to-case links	No	Yes
`heuristics`	10	Manually seeded operating laws	Yes — manually curated	Partially — seed.py has them hardcoded
`embeddings`	579	768d float32 Gemini vectors as blobs	Yes	Yes — but costs Gemini API calls (~11 min at free tier)
`edges`	44	Causal relationships between entities	No	Yes
`raw_pieces_summaries`	526	Raw PiecesOS annotation text cache	Yes — source of truth	Only if PiecesOS still has the data
`raw_import_batches`	11	Import run metadata	No	Yes
`raw_pieces_events`	0	Clipboard/vision/audio events (never populated)	—	—
`context_metrics`	207	Session scoring data (all collapse_score=0)	No	Yes
`briefings`	0	Never written — briefings print only	—	—
`interventions`	0	Never written — collapse detector produces 0	—	—
`tags`	0	Unused	—	—

Critical Data Integrity Facts

- **NULL vs "unknown"**: 311 entities have `project_id = NULL` (SQL null). The string `"unknown"` is a projects row but is NEVER stored in entities. Prior documentation was wrong.
- **8 cases have NULL project_id**: Failure chains built from NULL-project events inherit NULL. This is correct SQL behavior — FK allows NULL.
- **Embeddings stored as raw float32 blobs**: 768 floats × 4 bytes = 3072 bytes each. Verified: `vec_bytes=3072`. Use `struct.pack/unpack` (in `gemini_embed.py`) to decode.
- **Schema drift risk**: If `SCHEMA_SQL` changes and DB already exists, `init_db()` uses `CREATE TABLE IF NOT EXISTS` — new columns won't be added to existing tables. Must use ALTER TABLE migrations.

What Must Be Backed Up Before Any Changes

1. `meops.db` — the entire DB file
2. `core/heuristics/seed.py` — manually crafted heuristics (precious if modified)
3. `core/ledger/classify.py` — keyword signals (precious if tuned)

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10. CONFIRMED WORKING / PARTIAL / BROKEN / STUBBED

✅ Confirmed Working (Direct Evidence)

- Full pipeline: `python run_phase2.py full` — runs end-to-end, verified 2026-03-21
- PiecesOS MCP ingestion: 526 summaries fetched and stored
- Session building: 207 sessions from 526 events
- Failure lifting: 26 failures from event text
- Outcome lifting: 9 outcomes from event text
- Failure chain building: 9 chains
- Win signature building: 9 signatures
- Gemini embedding: 579 entities at 768d, confirmed 3072-byte blobs
- Cosine semantic recall: similarity 0.74–0.76 on IronClad queries
- TF-IDF fallback: triggered on 429, returns results
- Daily briefing: template output with live DB stats
- `run_phase2.py stats`, `recall`, `brief`, `embed` commands all work
- `GEMINI_API_KEY` env override of hardcoded key works correctly

⚠️ Partially Working

- **Project classification**: 215/526 events (41%) correctly classified. 311 NULL. Functional but low coverage.
- **Failure detection**: 26/526 (5%) — likely higher in reality but detection signals still miss many narrative-style failures
- **Context collapse detector**: Runs without error, scores sessions, writes to context_metrics — but threshold never reached due to broken scoring input
- **Briefings**: Output is real and readable — but narrative is hardcoded, not derived from actual case analysis

❌ Broken / Dead

- `pieces_bridge/ingest.py`: Appears active, is dead. REST endpoint returns 404. Do not use.
- **Decision extraction**: `decision_replay.py` produces 0 decisions from 526 events
- **Context collapse alerts**: 0 interventions from 207 sessions — scoring logic broken
- **WebSocket real-time streaming**: Code exists in `ingest.py` but file is dead
- `raw_pieces_events` **table**: 0 rows. Never populated. Clipboard/vision/audio workstream events not ingested.

⬜ Stubbed / Placeholder

- `core/causal/`: Empty. Phase 4 causal inference not started
- `pipelines/`: Empty. Standalone pipeline scripts not built
- `core/models/*.py`: Pydantic models defined but not validated on writes — reference only
- `core/briefing/daily.py` **narrative**: Hardcoded text — not ML-derived
- **Heuristic auto-mining (Phase 4)**: Not built. 10 heuristics are manual seeds only
- **Live operator guardrails (Phase 5)**: Not built

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11. SECURITY AND OPERATIONAL RISKS

#	Risk	Severity	What Could Happen	Location	Immediate Mitigation
1	**Two Gemini API keys in source**	🔴 CRITICAL	Key exposure if code shared, committed to git, or viewed by unauthorized party. API costs charged to account.	`run_phase2.py:6`, `gemini_embed.py:17`, `_test_embed.py:6`	Rotate both keys NOW. Set via env var.
2	**No DB backup**	🔴 CRITICAL	Single accidental delete or schema migration error = 526 events, 579 embeddings, 207 sessions lost. Embeddings cost ~11 min to rebuild at free tier.	`meops.db`	`copy meops.db meops.db.bak.YYYYMMDI` now
3	`_reset_canonical.py` **is silent + destructive**	🔴 HIGH	Running it accidentally wipes all derived intelligence. No confirmation prompt. Labeled "reset" not "DESTROY."	`_reset_canonical.py`	Add `input("Type DESTROY to confirm: ")` guard
4	`ingest.py` **looks active but is dead**	🟡 MEDIUM	New developer imports and calls it, gets 404 silently on all items, thinks	`pieces_bridge/ingest.py`	Add `raise RuntimeError("DEAD — use ingest_p2.py")` at top

			ingestion worked		
5	**No test suite**	🟡 MEDIUM	Any refactor risks silent regression. Classification changes, schema changes, embedding changes all unverifiable without running full pipeline.	Entire project	None
6	**Port 39300 hardcoded**	🟡 MEDIUM	PiecesOS port varies per install. Ingestion silently fails on different machines.	`mcp_client.py:15`, `ingest.py:24-25`	Check PiecesOS Quick Menu for actual port
7	`requirements.txt` **missing packages**	🟡 MEDIUM	Fresh install succeeds, pipeline fails at runtime: `google.genai` not found, `numpy` not found.	`requirements.txt`	Fix immediately (see Section 5)
8	**PYTHONIOENCODING not set globally**	🟢 LOW	Any script that prints non-ASCII on Windows crashes. `run_phase2.py` sets it but other scripts don't.	All scripts	Run with `set PYTHONIOENCODING=utf8 &&` prefix
9	**No `.gitignore`**	🟢 LOW	If project is ever committed to git: API keys, DB file, audit scripts, scratch files	Root	Create `.gitignore` immediately

			all get committed		

12. MAINTENANCE GUIDANCE

Where to Start

1. `python run_phase2.py stats` — confirm current DB state
2. `python run_phase2.py recall "test query"` — confirm recall working
3. Read `core/ledger/classify.py` — most impactful single file for data quality
4. Read `core/recall/gemini_embed.py` — most complex module; controls Phase 3

Do NOT Touch First

- `core/storage/schema.py` — any column change requires migration + potential data wipe
- `core/storage/db.py` → `EntityRepo.upsert()` — load-bearing for all ingestion
- `pieces_bridge/mcp_client.py` — depends on exact PiecesOS MCP protocol version

Load-Bearing Files (require backup + test before changing)

1. `core/storage/db.py`
2. `core/storage/schema.py`
3. `pieces_bridge/mcp_client.py`
4. `pieces_bridge/ingest_p2.py`
5. `core/recall/gemini_embed.py` (blob format — changing breaks all embeddings)

Safe to Iterate Freely

- `core/ledger/classify.py` — add keywords
- `core/heuristics/seed.py` — add heuristics
- `core/briefing/daily.py` — improve templates
- `run_phase2.py` — add new CLI commands

Before Any Schema Change

```
copy meops.db meops.db.bak.YYYYMMDD
# Write migration as ALTER TABLE in _migrate_vN.py
# Test on fresh DB: set MEOPS_DB=meops_test.db && python run_phase2.py full
```

Incremental Improvement Path (Safe)

1. Add keywords to `classify.py` → run `python run_phase2.py full` → check `stats`
2. Add heuristics to `seed.py` → run `init_db()` → verify in DB
3. Improve briefing templates in `daily.py` → no DB impact
4. Tune failure signals in `classify.py` → rebuild cases

---

13. EDGE CASES AND HIDDEN FAILURE MODES

#	Edge Case	Type	Evidence	Impact
1	`ingest.py` **looks primary but is dead**	Dead code masquerading as active	REST 404 on all IDs, confirmed	HIGH — misleads new developer
2	**NULL vs "unknown" project_id**	Documentation error	Audit confirms 311 NULL, 0 "unknown"	MEDIUM — querie for project_id='unknov return empty
3	**Two different API keys**	Configuration conflict	Audit confirms two distinct keys in env vs source	HIGH — unpredictable behavior on fresh machine without er set
4	**Embeddings fail silently after 100 RPM**	Quota + retry logic	429 caught, retried, eventually stored partial results	MEDIUM — 461/56 may not be embedded after rat limit. Always check `embeddings` count after `embed` run.
5	**PYTHONIOENCODING not set = crash**	Windows encoding	Any script printing → or ✅ without utf-8	MEDIUM — only `run_phase2.py` sets i
6	`tool_switch_count` **is not real switches**	Wrong implementation	All sessions show switch_count 0-1	MEDIUM — collaps detector useless
7	`raw_pieces_summaries` **= 526 not 1374**	Prior doc was wrong	Confirmed by audit	MEDIUM — prior handoff document wrong
8	**Pydantic models not validated on write**	Silent schema bypass	`EntityRepo.upsert()` accepts raw dicts	LOW — type safety absent
9	`_reset_canonical.py` **has no guard**	Destructive script	Code inspection	HIGH — easy accidental data loss
10	`context_metrics.fragmentation_score` **added post-deployment**	Schema drift	`_migrate.py` evidence	LOW — fixed, but reveals schema management is ad-hoc
11	**PiecesOS port varies per install**	Machine coupling	39300 hardcoded in 3 files	MEDIUM — breaks on non-standard install

12	`tmp/pdfs/generate_me_ops_app_summary.py`	Orphaned file	Not imported anywhere	LOW — confusing
13	`run.py` **still exists alongside** `run_phase2.py`	Dual entry points	Both present, both importable	LOW — confusion about which to use

---

14. UPGRADE AND STABILIZATION ROADMAP

Phase 0 — Secure and Stabilize (Do Today)

Task	Why	Effort	Risk
Rotate + move both Gemini API keys to `.env`	Keys exposed in source	15 min	Low
Backup `meops.db`	No backup exists	5 min	None
Fix `requirements.txt`	Fresh installs fail	5 min	None
Add guard to `_reset_canonical.py`	Silent data destruction	5 min	None
Mark `ingest.py` as dead	Misleads maintainers	5 min	None
Delete scratch scripts	Clutter	5 min	None
Create `.gitignore`	Prevent secret/DB commit	5 min	None

Phase 1 — Reproducible MVP

Task	Why	Effort	Payoff
Add `python-dotenv` + `.env` loading	Proper secrets management	30 min	High
Write 3 unit tests for `classify.py`	Fragile — changing breaks classification silently	2 hrs	High
Fix `decision_replay.py` using embedding similarity	Keyword triggers don't work	4 hrs	Medium
Fix `collapse_detector.py` using real app-switch data	Current logic always produces 0	4 hrs	Medium

Phase 2 — Developer Ergonomics

Task	Why	Effort	Payoff
Improve `classify.py` for 311 unknowns	59% miss rate	2 hrs	High
Add logging module (replace print)	No observability	2 hrs	Medium
Schema migration system (Alembic lite)	Ad-hoc ALTER TABLE is fragile	4 hrs	Medium
Consolidate `run.py` + `run_phase2.py` into single CLI	Dual entry points cause confusion	2 hrs	Low

Phase 3 — Data Quality

Task	Why	Effort	Payoff
Ingest `raw_pieces_events` (clipboard/vision/audio)	Currently 0 rows — rich signal available	8 hrs	High

Improve failure signal detection	26/526 (5%) is too low	3 hrs	High
Upgrade briefings to use actual case analysis	Currently hardcoded narrative	4 hrs	High
Add context collapse scoring from real app data	Currently always 0	4 hrs	Medium

Phase 4 — Higher-Order Intelligence

Task	Why	Effort	Payoff
Build `core/heuristics/mine_rules.py`	Contrastive win/failure analysis → auto-generated operating laws	16 hrs	Very High
Build `core/causal/` inference module	Currently empty scaffold	20 hrs	High
Upgrade decision extraction to embedding-based	Keyword approach fundamentally broken	8 hrs	High

Phase 5 — Agentification

Task	Why	Effort	Payoff
OpenClaw MCP integration	ME-OPS as tool for autonomous agent	8 hrs	High
Paperclip CEO agent feed	ME-OPS feeds governance layer	8 hrs	High
Live guardrails + interrupt system	Detect dead-end patterns in real time	20 hrs	Very High

---

15. CHANGE-CONTROL RULES

1. **Back up `meops.db` before any schema change, migration, or reset operation**
2. **Never call `pieces_bridge/ingest.py` — it is dead and produces bad data**
3. **Never edit `meops.db` directly in SQLite browser without a backup**
4. **Add tests before refactoring `classify.py`, `failure_chain.py`, or `gemini_embed.py`**
5. **Never commit API keys — use env vars and `.env` file**
6. **Trust runtime behavior over documentation** — this document supersedes all previous handoff docs
7. **After any `run_phase2.py full` run, verify with `stats` command** — check embedding count matches entity count
8. **When adding new project buckets to `classify.py`**: add to `SEED_PROJECTS_SQL` in `schema.py` too
9. **When changing `EntityRepo.upsert()` signature**: all ingestion breaks — test with small dataset first
10. **Do not use PowerShell `-c` for multi-line Python** — use `.py` files

---

16. OPEN QUESTIONS

Must Answer Now

1. **Which Gemini API key is the real/intended one?** Two different keys in use. After rotation, ensure only one key in `.env`.
2. **Is the current DB the full history or a partial subset?** `raw_pieces_summaries` has 526 rows; PiecesOS MCP returns 1000 IDs. 474 IDs returned no annotation text. Are those summaries genuinely empty or is

there a fetch path issue?

Should Answer Soon

3. **Should `gettupp-ent` and `gettupp-girls` appear in classify signals?** Currently in `PROJECT_SIGNALS` but no events classified to those projects. Is there really no GettUpp-specific activity in captured history?
4. **What are the 311 NULL-project events?** Run: `SELECT summary FROM entities WHERE project_id IS NULL LIMIT 20` — inspect to find the dominant pattern and add signals.
5. **Is `ingest.py` safe to delete?** Confirmed dead but some imports may reference it. Verify: `findstr /s "from pieces_bridge.ingest import" *.py`

Can Defer

6. Should ME-OPS eventually sync to cloud / be accessible remotely?
7. Should the Paperclip CEO agent spec (captured in clipboard) be formalized?
8. Is the airbnb cleaning GTM from March 20 still relevant?

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17. IMMEDIATE NEXT ACTIONS (PRIORITY ORDER)

Do today — security and data safety:

```
1. [ ] Rotate both Gemini API keys (AIzaSyBcXk... and AIzaSyAnLC...)
2. [ ] copy meops.db meops.db.bak.20260321
3. [ ] Create .env file with GEMINI_API_KEY=new_key
4. [ ] Fix requirements.txt — add google-genai>=1.68.0 and numpy>=2.0
5. [ ] Add "DEAD — use ingest_p2.py" to top of ingest.py
6. [ ] Add input() guard to _reset_canonical.py
```

Do before any feature work:

```
7. [ ] Create .gitignore (exclude: meops.db, .env, _*.py, *.bak, __pycache__, *.pyc)
8. [ ] Add python-dotenv to requirements.txt and load .env in run_phase2.py
9. [ ] Run: SELECT summary FROM entities WHERE project_id IS NULL LIMIT 30 — identify top
unknown patterns
10.[ ] Add 10+ keywords to classify.py for the dominant unknown session types
11.[ ] python run_phase2.py full — verify classification improves
```

Fix broken modules (before Phase 4):

```
12.[ ] Fix decision_replay.py — replace keyword triggers with embedding similarity
13.[ ] Fix collapse_detector.py — add real app-switch signal from workstream_events
14.[ ] Write basic tests for classify.py (at minimum: known project text → expected
project_id)
```

Clean up:

```
15.[ ] Delete: _debug.py, _enc.py, _migrate.py, _audit.py, _audit2.py, _audit3.py
16.[ ] Decide: merge run.py into run_phase2.py or delete run.py
17.[ ] Move tmp/pdfs/generate_me_ops_app_summary.py to docs/ or delete
```

Forensic audit completed: 2026-03-21 | Evidence base: live DB query, file inspection, import tests, API health checks, runtime verification | All major claims labeled Confirmed, Likely, Unknown, or Conflicting